

Bingran Li

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SUMMARY

From Humanities high school student to Mathematics undergraduate interested in theory and application of optimization and probability. Trying theory into practice, especially OR and AI

My hobby is learning any math and taking **Math for AI** and **AI for Math** as an ultimate dream.

EDUCATION

The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

Bachelor of Science in Mathematics and Applied Mathematics (Expected Graduation: 2026)

Sept. 2022 – Present

- Cumulative GPA: 3.919/4.000 — Rank: **1st/74**
- Honors: National Scholarship for Encouragement(2023); Undergraduate Research Awards(2024, 2025); Academic Performance Scholarship(2022-2023, 2023-2024); Muse College Scholarship(2022-2024)
- **Postgraduate Courses:** Topics in Financial Mathematics, Measure Theoretic Probability, Numerical Linear Algebra, Optimization in Machine Learning, Advanced Convex Optimization, Advanced Stochastic Processes (Audited)
- **Specialization:** Modern Optimization (Prof. Yurii Nesterov) , Optimization in Machine Learning and Data Science (Prof. Yinyu Ye)

The Chinese University of Hong Kong

Hong Kong SAR, China

International Summer School

June 2024 – August 2024

SKILLS

Frameworks: PyTorch

Languages: Python, C/C++, MATLAB

Libraries: NumPy, Pandas, Matplotlib

Languages: English (Fluent), Mandarin (Native)

RESEARCH

Stochastic Control for Portfolio Selection

Oct. 2023 – Jul. 2024

with Prof. Yang Liu

- Developed a closed-form optimal strategy for Merton's problem under market-based Value-at-Risk constraints, which is a stochastic optimal control problem in a non-convex setting

Concentration Inequality in Reinforcement Learning

Aug. 2024 – Oct. 2024

with Mr. Yingru Li

- Write some examples of one-hot feature multi-armed bandit using a new probability tool developed by a Postgraduate student

Course Project (DDA6202 Optimization in Machine Learning)

Fall 2024

Title: Analyzing Non-diagonal Preconditioning Optimizers in Deep Learning

- Analyze and compare the performance between diagonal preconditioning optimizers (e.g. Adam-mini) and non-diagonal preconditioning optimisers (e.g. SOAP) on some simple models, such as linear model, MLP, and Transformer to analyze their pros and cons.

Algorithmic Complexity in Markov Decision Process

Oct. 2024 – Present

Strongly Polynomial-Time Policy Iteration

with Prof. Yinyu Ye

- Prove the strongly polynomial time for the policy iteration method in deterministic Markov decision processes

Stochastic Optimization in Deep Learning

Feb. 2025 – Present

Convergence results of Adam

with Prof. Ruoyu Sun & Mr. Yushun Zhang

- Prove that Adam under with-replacement sampling with $\beta_1 > 0$ can converge when β_2 is large enough

Implementation of an Interior Point method(IPM)

April. 2025 – Present

with Prof. Yin Zhang & Prof. Yurri Nesterov

- Implement the Asymmetric Long-Step Primal-Dual IPMs with Dual Centering, on Linear Programming and SDP problems

READING GROUPS

Concentration Inequalities

Jan. 2024 – Jul. 2024

with Prof. Jeff Yao

- Gave 3 lectures on the Johnson-Lindenstrauss Lemma and Transportation Method

Graph Theory

Jan. 2025 – Present

with Prof. Wingki Shum

- Give lectures based on a textbook

Analytic Number Theory

Jan. 2025 – Present

with Prof. Tiefeng Jiang

- Give lectures on Arithmetic Function Theory and explore proving Riemann Hypothesis

Data Science

Feb. 2025 – Present

textbook: Foundations of Data Science

with Prof. Zhi-Quan Luo

- Give presentations on the textbook: high dimensional probability and geometry
- Report on the maximum number of almost orthogonal unit vectors in $\mathbb{R}^d$

TEACHING

Lecturer

Sept. 2023 – Present

Academic Department of Statistics Alliance

- MAT2050 Mathematical Analysis Midterm Review, 2023-2024 Fall term
- MAT2004 Mathematical Statistics Midterm Review, 2023-2024 Spring term

Teaching Fellow

March 2024 – May 2024

School of Science and Engineering Help Room

- Held 5-hours office hours every week to tutor students

Undergraduate Student Teaching Fellow

Sept. 2024 – Present

MAT2001 Honours Ordinary Differential Equations

- Gave one-hour lectures, graded assignments, and held a one-hour office hour weekly

STA4001 Stochastic Process

- Give 4 one-hour lectures for around 40 students in tutorial sessions

Name: LI, Bingran Chinese Name: 李炳然
Student ID No.: 122020087 ID/Passport No.: 420106200408128410 Date of Birth: 12 Aug

Admitted in: Sep 2022
College: Muse College
School: School of Science and Engineering
Major/Programme: Mathematics and Applied Mathematics
Stream(s): Financial Mathematics
Mode of Study: Full-time

2022-23 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
ACT2111	Introductory Financial Accounting	3.0	A	37.0
CSC1001	Introduction to Computer Science: Programming Methodology	3.0	A	38.8
ECO2011	Basic Microeconomics	3.0	A	34.3
ENG1001	English Bridge Program (EBP)	3.0	B+	31.2
GEW1001	Ethics and the Rule of Law	3.0	PA	N/A
ITE1000	Information Technology	1.0	PA	N/A
MAT1001	Calculus I	3.0	A	33.4
PED1001	Physical Education	1.0	B	38.3
		Units Passed = 20.0	Term GPA = 3.806	
		Cumulative Units Passed = 20.0	Cumulative GPA = 3.806	

2022-23 Term 2

Course Code	Course Title	Units	Grade	% of A- and above
CHI1000	Chinese	3.0	A-	37.3
ECO2121	Basic Statistics	3.0	A	39.4
ENG1002	English for Academic Purposes I	3.0	A-	34.5
FIN2010	Financial Management	3.0	A	39.6
GEW2001	Introduction to Marxism	3.0	DI	N/A
MAT1002	Calculus II	3.0	A	41.4
PED1002	Fitness and Health	1.0	A	38.5
		Units Passed = 19.0	Term GPA = 3.888	
		Cumulative Units Passed = 39.0	Cumulative GPA = 3.847	

Honour(s)/Award(s): Dean's List 2022-23

2022-23 Summer Session

Course Code	Course Title	Units	Grade	% of A- and above
MAT2040	Linear Algebra	3.0	A	39.1
		Units Passed = 3.0	Term GPA = 4.000	
		Cumulative Units Passed = 42.0	Cumulative GPA = 3.860	

2023-24 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
ECO3121	Introductory Econometrics	3.0	A	41.5
GFH1000	In Dialogue with Humanity	3.0	A	37.2
MAT2001	Honours Ordinary Differential Equations	3.0	A	38.8
MAT2050	Mathematical Analysis	3.0	A	38.7
MAT3007	Optimization	3.0	A	37.9

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2023-24 Term 1

MAT3040	Advanced Linear Algebra	3.0	A-	33.6
STA2003	Probability	3.0	A	37.5
		Units Passed = 21.0	Term GPA = 3.957	
		Cumulative Units Passed = 63.0	Cumulative GPA = 3.896	

2023-24 Term 2

Course Code	Course Title	Units	Grade	% of A- and above
ERG2081	Independent Study I	2.0	A	96.8
FIN3080	Investment Analysis and Portfolio Management	3.0	A	39.2
GFN1000	In Dialogue with Nature	3.0	A	38.7
MAT4001	Numerical Analysis	3.0	A-	39.0
MAT7710	Topics in Financial Mathematics	3.0	A	71.4
STA2004	Mathematical Statistics	3.0	A	39.8
STA4001	Stochastic Processes	3.0	A	40.5
		Units Passed = 20.0	Term GPA = 3.955	
		Cumulative Units Passed = 83.0	Cumulative GPA = 3.912	

Honour(s)/Award(s): Dean's List 2023-24

2023-24 Summer Session

Course Code	Course Title	Units	Grade	% of A- and above
DDA4330	Modern Optimization	1.0	P	N/A
		Units Passed = 1.0	Term GPA = 0.000	
		Cumulative Units Passed = 84.0	Cumulative GPA = 3.912	

2024-25 Term 1

Course Code	Course Title	Units	Grade	% of A- and above
DDA6020	Measure Theoretic Probability	3.0	A-	52.9
DDA6202	Optimization Models and Methods in Machine Learning	3.0	A	64.1
DDA6204	Selected Topics in Data and Decision Analytics IV	3.0	A	50.0
GEC2206	Seminar on Selected Topics of Global Political Economy	3.0	A	31.7
MAT3006	Real Analysis	3.0	A	39.5
MAT4220	Partial Differential Equations	3.0	A	35.4
		Units Passed = 18.0	Term GPA = 3.950	
		Cumulative Units Passed = 102.0	Cumulative GPA = 3.919	

2024-25 Term 2

Course Code	Course Title	Units	Grade	% of A- and above
DDA4300	Optimization in Data Science and Machine Learning	1.0	IP	0.0
DDA6110	Advanced Convex Optimization	3.0	IP	0.0
ERG2082	Independent Study II	2.0	IP	0.0
MAT4010	Functional Analysis	3.0	IP	0.0
		Units Passed = 0.0	Term GPA = 0.000	
		Cumulative Units Passed = 102.0	Cumulative GPA = 3.919	

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***** Summary *****			
	Total Units Passed	Cumulative GPA	Major GPA
CUHK(SZ)	102.0	3.919/4.000	3.960/4.000
Unit Exemption	6.0	-	-

Remarks:

6.0 units of course(s) taken at another university recognized as equivalent.

Major GPA is the grade point average across a list of Major courses defined by the School Board, as being most relevant to the student's Major.

***** End of Transcript *****

Director of Registry Services